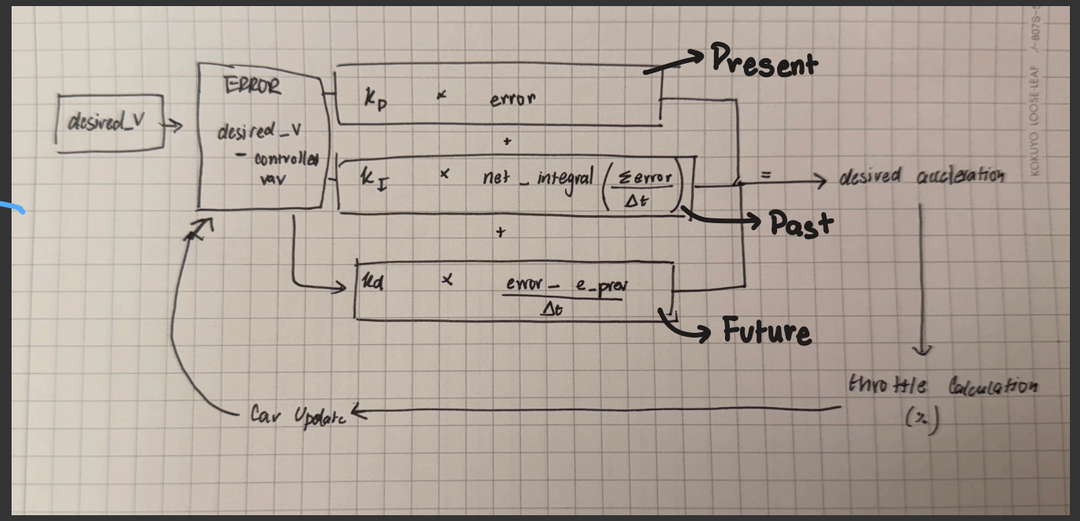
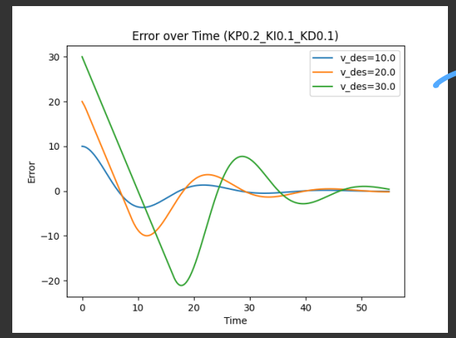
Formula Slugs '26

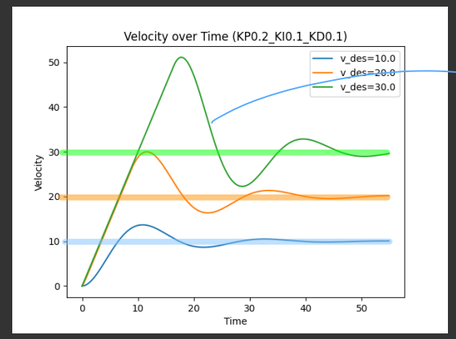


PID Controller

> Control of the velocity of the car



→ with a full PID



Overshooting at first then undershooting

↓

Dynamic weights ?

VS

CONTROLLER FOR VELOCITY



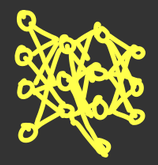
– By Ashmita Dua

PID vs MLP Comparison (K_P = 0.2, K_I = 0.1, K_D = 0.1)

controller	desired_v	final v	final error	overshoot
PID	10.0	10.066	-0.066	36.7%
PID	20.0	20.180	-0.180	49.9%
PID	30.0	29.594	0.406	70.3%
MLP	10.0	10.054	-0.054	36.4%
MLP	20.0	20.166	-0.166	49.6%
MLP	30.0	29.600	0.400	70.3%

↳ Slightly better results

MLP (Multi-layer Preceptron)

Velocity, desired_velocity →  → throttle percentage (-1, 1)

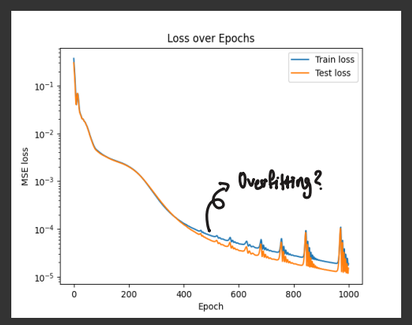
> 3 linear layers

2 → 16 → 16 → 1

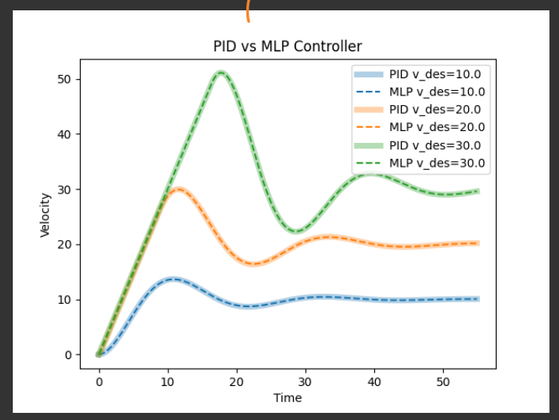
> RELU activation

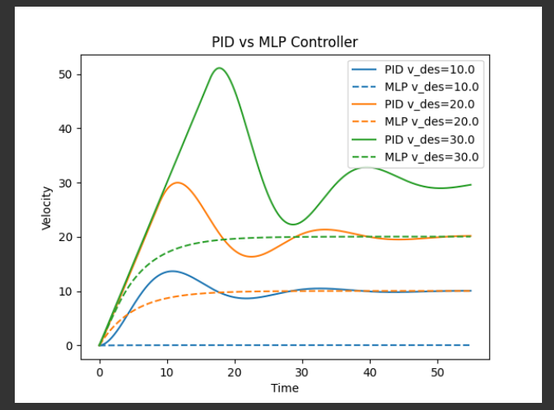
> 1000 Epochs over 4000 training rows

↳ on synthetic data produced from PID



full PID data





⇒ Synthetic data generated by a PI

Somehow trained the model better !!

↳ MORE CONSISTENT RUN